CALCULUS OF INVERSE FUNCTIONS QUIZ

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Name: Solutions

Differentiation

Compute the derivative of the following functions.

1. $\ln(\cos(x))$

$$\frac{d}{dx}\ln(\cos(x)) = \frac{\frac{d}{dx}\cos(x)}{\cos(x)}$$

$$= \frac{-\sin(x)}{\cos(x)}$$

$$= \boxed{-\tan(x)}$$

2. $e^{\sqrt{x}}$

$$\frac{d}{dx}e^{\sqrt{x}} = e^{\sqrt{x}}\frac{d}{dx}\sqrt{x}$$

$$= e^{\sqrt{x}}\frac{1}{2}x^{-1/2}$$

$$= \frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

Integration

Evaluate the following indefinite integrals.

3. $\int \tan(x)\ln(\cos(x))\,\mathrm{d}x$

$u = \ln(\cos(x))$

$du = -\tan(x)\,dx$

$\Rightarrow -du = \tan(x)\,dx$

$$\int \tan(x)\ln(\cos(x))\,dx = -\int u\,du$$

$$= -\frac{1}{2}u^2 + C$$

$$= \boxed{-\frac{1}{2}\ln(\cos(x))^2 + C}$$

4. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\,\mathrm{d}x$

$u = e^{\sqrt{x}}$

$du = \frac{e^{\sqrt{x}}}{2\sqrt{x}}dx$

$\Rightarrow 2du = \frac{e^{\sqrt{x}}}{\sqrt{x}}dx$

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}}dx = \int 2u\,du$$

$$= 2\left(\frac{1}{2}u^2\right) + C$$

$$= \left(e^{\sqrt{x}}\right)^2 + C$$

$$= \boxed{e^{2\sqrt{x}} + C}$$